

1 Highlights

2 Preictal Mutual Information Dynamics Bifurcate by Seizure Onset Lobe: A Zero-Parameter, Four-Measure Information-
3 Theoretic Analysis of the HUP iEEG Cohort

4 Arif Faysal Nayem

5 • Uniform preictal MI collapse hypothesis rejected ($p =$
6 0.92 , $N = 54$ patients).

7 • Preictal MI bifurcates by lobe: collapse in frontal-
8 onset ($d \approx +0.99$), amplification in MTL-onset ($d \approx$
9 -2.63).

10 • Bifurcation confirmed independently by graph topol-
11 ogy and MI-variance divergence across three mea-
12 sures.

13 • 6 of 16 frontal-onset patients show an invisible pre-
14 cursor: MI collapses while signal variance increases
15 simultaneously.

16 • Mixed-cohort pooling mechanically produces a null re-
17 sult, resolving two decades of contradictory literature.

Preictal Mutual Information Dynamics Bifurcate by Seizure Onset Lobe: A Zero-Parameter, Four-Measure Information-Theoretic Analysis of the HUP iEEG Cohort

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Abstract

Mutual information (MI) has been proposed as a preictal biomarker for focal epilepsy, yet published studies have reported opposite directions of effect: some observe that network-wide MI decreases before seizure onset while others observe that it increases, with no mechanistic account of the inconsistency. We tested the uniform MI collapse hypothesis — that global mean pairwise MI decreases preictally across all patients — in 54 patients from the Hospital of the University of Pennsylvania (HUP) intracranial EEG (iEEG) dataset using four sequential, zero-parameter information-theoretic measures: rolling Kraskov-estimated global MI, MI-weighted dynamic network graph topology, MI versus signal variance, and Transfer Entropy (TE) directional asymmetry. The uniform hypothesis was rejected (Wilcoxon signed-rank, $p = 0.92$), and patient-level analysis revealed that the rejection arose from two equally prevalent, biologically opposed dynamics: frontal-onset patients exhibited preictal MI collapse (Cohen’s $d \approx +0.99$), while mesial temporal lobe (MTL)-onset patients exhibited preictal MI amplification ($d \approx -2.63$). This lobe-specific bifurcation was confirmed independently by graph topology (16 fragmentation vs. 15 integration among 31 non-saturated patients; Spearman $r = -0.60$ between MI and network density) and by MI-variance divergence (6 of 16 frontal-onset patients showed MI collapse while signal variance simultaneously increased; frontal-onset interictal $r(\Phi, V) = 0.362$ vs. MTL-onset $r = 0.497$). Preictal MI dynamics are lobe-specific rather than universal; mixed-cohort studies that pool both phenotypes will observe a null result — explaining two decades of contradictory findings — and the MI directionality phenotype constitutes a parameter-free biomarker of seizure network type, calibrated from the interictal baseline without any labelled seizure data.

Keywords: seizure prediction, mutual information, intracranial EEG, information bottleneck, network bifurcation, graph fragmentation, Transfer Entropy, zero-parameter biomarker, preictal dynamics

1. Introduction

Drug-resistant focal epilepsy affects approximately 20 million people worldwide, and for these patients surgical resection of the seizure onset zone (SOZ) and closed-loop responsive neurostimulation are the primary treatment options [1, 2]. Both applications require preictal biomarkers derived from iEEG recordings without trained classifiers: SOZ localization demands biomarkers that rank ictal-driving channels above healthy tissue, while closed-loop stimulation demands biomarkers that detect the preictal transition before the electrographic discharge propagates [3]. Mutual information (MI) — a nonlinear, model-free measure of statistical dependence between iEEG channels — has been proposed as such a biomarker [4, 5], but published studies have reported opposite directions of preictal change. Without an account of why this inconsistency arises, no reliable MI-based detection rule

can be constructed. We tested the uniform MI collapse hypothesis (global mean pairwise MI decreases before all seizures relative to interictal baseline) in 54 patients from the HUP iEEG dataset [6] using four sequential, zero-parameter information-theoretic measures, and characterised the patient-level structure underlying the departure from that hypothesis.

Prior work falls into three thematic lines, each with a direct limitation for the present question. Synchrony and MI as group-level biomarkers. Mormann et al. (2007) [3] reviewed two decades of preictal synchrony evidence, concluding that MI decreases are observed predominantly in temporal lobe epilepsy cohorts while frontal and extratemporal cohorts show inconsistent effects; no mechanistic explanation of the inconsistency was proposed. Schindler et al. (2008) [4] demonstrated measurable changes in multichannel correlation structure at seizure onset, but their measure is linear and symmetric, capturing co-variation without directionality. Schreiber et al. (2025) [7] identified transient high-connectivity epochs in the hours preceding seizures, but their measure operates

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on hour-long timescales and uses linear coherence rather than full statistical dependence. None of these works characterised the directional inconsistency as a systematic patient-level bifurcation tied to lobe of onset. Graph theoretic network analysis. Kramer and Cash (2012) [5] framed epilepsy as a network disorder and reviewed lobe-specific topology changes at ictal onset, but used linear correlation matrices rather than MI-weighted graphs and did not compare systematically defined patient subgroups. Bullmore and Sporns (2009) [8] established the brain-as-graph framework and showed that healthy networks exhibit small-world topology that may be disrupted during pathological transitions — motivating network density, degree, and clustering as preictal metrics — but these measures cannot distinguish between the fragmentation predicted by information bottleneck theory and the integration predicted by hypersynchrony theory. Transfer Entropy for SOZ identification. Zhao et al. (2024) [9] achieved 97.24% seizure classification accuracy by feeding TE values into a trained convolutional network, but this embeds TE inside a black-box classifier rather than using it as a standalone zero-parameter measure. Concetti et al. (2024) [10] outperformed undirected connectivity measures for SOZ localization using causal network analysis, but required a labelled training set. Van Mierlo et al. (2013) [11] demonstrated that phase TE identifies SOZ channels in temporal lobe SEEG recordings, but did not evaluate across mixed lobe types or without training. No prior work derived a patient-agnostic SOZ ranking from TE directional asymmetry alone.

The gap across all three lines is the absence of a systematic account of why preictal MI direction differs by patient. No existing study has tested the uniform collapse hypothesis on a large public cohort, observed its rejection, and characterised the patient-level structure of that rejection in terms of lobe-specific ictogenic mechanisms. We address this gap by showing that the directionality of preictal MI change is a stable, lobe-specific patient-level phenotype, confirmed independently by three information theoretic measures. Our specific contributions are:

- A patient-level bifurcation of preictal MI dynamics. We show that the uniform MI collapse hypothesis is rejected ($p = 0.92$) because frontal-onset patients exhibit collapse (Cohen’s $d \approx +0.99$; frontal SOZ enrichment OR = 7.54) while MTL-onset patients exhibit amplification ($d \approx -2.63$; MTL depletion OR = 0.26) with both directions predicted by distinct ictogenesis mechanisms.
- Convergent three-measure validation. We confirm the bifurcation independently via MI-weighted graph topology (16 fragmentation / 15 integration among 31 non-saturated patients; $r = -0.60$ between MI and network density) and via MI-variance divergence (frontal-onset interictal $r = 0.362$ vs. MTL-onset $r = 0.497$) providing convergent validity that the bifurcation is a real network phenomenon.

- An invisible preictal precursor in frontal-onset patients. We identify that 6 of 16 frontal-onset patients exhibit MI collapse while signal variance simultaneously increases — a preictal network reorganisation undetectable by standard amplitude-based clinical alarms.
- A resolution of the prior literature inconsistency. We demonstrate that pooling frontal-onset and MTL-onset patients mechanically produces a null result, because the two phenotypes cancel in a signed-rank test, explaining why MI-based seizure studies have reported contradictory findings for two decades.

Section 2 describes the dataset and preprocessing. Section 3 defines the four information-theoretic measures. Section 4 reports the experimental results. Section 5 discusses biological interpretation, clinical implications, and limitations. Section 6 concludes.

2. Dataset and Preprocessing

2.1. HUP iEEG Dataset

We used the Hospital of the University of Pennsylvania (HUP) iEEG dataset [6], publicly available via NE-MAR (OpenNeuro ID: ds004100) and Pennsieve Discover (dataset 179) in Brain Imaging Data Structure (BIDS) format. The dataset comprises de-identified intracranial EEG recordings from patients with drug-resistant focal epilepsy who underwent presurgical iEEG monitoring. Electrode MNI coordinates appear in *_electrodes.tsv files; patient-level metadata — including seizure-onset lobe target (target), implant type (ECoG or SEEG), age, sex, and Engel outcome — appear in participants.tsv.

We processed 54 patients for whom both ictal and interictal recordings were available and passed quality checks (Section 2.3). One additional patient was included for Phase 4 only (55 total). The cohort comprises 85% ECoG (subdural grid and strip) and 15% SEEG (depth electrode) recordings, with channel counts ranging from 16 to 94 per patient. All analyses used the 16 channels of highest interictal variance per patient, enabling uniform computation across configurations.

Ictal recordings contain approximately 120 seconds of preictal data preceding the clinician-annotated seizure onset. Interictal recordings contain approximately 300 seconds recorded at least 4 hours from any annotated seizure. Seizure onset times were extracted from BIDS *_events.tsv files using the trial_type = "sz onset" label; files were read with the utf-8-sig codec to handle the UTF-8 byte-order mark present in HUP BIDS files.

2.2. Preprocessing

All preprocessing was applied identically across patients and epochs using NumPy, SciPy, and MNE-Python [12].

Bandpass filtering. A fourth-order zero-phase Butterworth filter (0.5–300 Hz) removed DC drift and electrode

drift artefacts while preserving high-frequency oscillations relevant to seizure dynamics.

Notch filtering. IIR notch filters at 60 Hz and harmonics (120 and 180 Hz, $Q = 30$) removed power-line interference.

Referencing. ECoG recordings used common average referencing (CAR), subtracting the mean across all channels to remove common-mode artefacts. SEEG recordings used bipolar referencing between adjacent contacts along each electrode shaft.

Artefact rejection. Channels with peak amplitude exceeding 3,000 μV (indicating amplifier saturation) were marked bad and excluded from MI computation. Patients with more than 20% bad channels were excluded from analysis.

Downsampling. All recordings were downsampled to 500 Hz before MI computation, preserving signal content to 250 Hz while reducing computation time by a factor of 2–4 relative to native sampling rates.

2.3. Quality Gating and Cohort Selection

Patients whose interictal mean MI fell below 0.02 nats — indicating signal dropout or near-zero variance channels — were excluded. Patients whose preictal mean MI exceeded 0.5 nats — indicating amplifier saturation artefacts producing spuriously large MI estimates — were excluded. These criteria yielded the 54-patient Phase 1 analysis cohort from an initial pool of 57.

2.4. MI Estimation

We estimated pairwise MI using the k -nearest-neighbour estimator of Kraskov, Stögbauer, and Grassberger (2004) [13]:

$$\hat{I}(X; Y) = \psi(k) - \langle \psi(n_x + 1) + \psi(n_y + 1) \rangle + \psi(N) \quad (1)$$

where ψ denotes the digamma function, $k = 5$ is the neighbour count, and n_x, n_y count neighbours within the per-sample radius defined by the k -th joint nearest neighbour. This estimator is asymptotically unbiased, requires no binning, adapts to local density variations, and its finite sample bias affects Φ_{inter} and Φ_{pre} equally — ensuring that relative changes are unaffected by the bias (see Appendix A).

MI was computed for all $\binom{16}{2} = 120$ channel pairs over a rolling window of 5 seconds (2,500 samples at 500 Hz), stepped at 1-second intervals (80% overlap). The implementation was validated against the minepy reference library on synthetic bivariate Gaussian data with known MI before application to iEEG data.

3. Methods

3.1. Notation

Let $\mathbf{x}(t) = \{x_1(t), \dots, x_N(t)\}$ denote the $N = 16$ -channel iEEG recording. Let t_s denote the clinician-annotated

seizure onset time. Let $x_i^{(w_t)}$ denote the segment of channel i within the 5-second rolling window centred at time t . Let μ_{inter} and σ_{inter} denote the mean and standard deviation of $\Phi(t)$ across the interictal epoch.

3.2. Phase 1: Global MI Signal and the Bifurcation Test

Global MI signal. Define the global mean pairwise MI at time t as:

$$\Phi(t) = \frac{2}{N(N-1)} \sum_i \sum_{j>i} \hat{I}(x_i^{(w_t)}; x_j^{(w_t)}) \quad (2)$$

$\Phi(t)$ is the mean over all 120 channel-pair MI values within window w_t . It is computed for every window step across both the interictal and preictal epochs.

Baseline statistics. From the interictal epoch, compute $\mu_{\text{inter}} = \text{mean}[\Phi]$ and $\sigma_{\text{inter}} = \text{std}[\Phi]$. From the final 60 seconds of the preictal epoch, compute $\mu_{\text{pre}} = \text{mean}[\Phi]$.

Effect size and phenotype classification. For each patient, compute Cohen’s d :

$$d = \frac{\mu_{\text{inter}} - \mu_{\text{pre}}}{s_{\text{pooled}}} \quad (3)$$

where s_{pooled} is the pooled standard deviation of the two epoch distributions. Positive d indicates MI decreases preictally (collapse, Type A); negative d indicates MI increases (amplification, Type B). Patients are classified as Type A if $d > 0.3$, Type B if $d < -0.3$, and indeterminate if $|d| \leq 0.3$.

Group hypothesis test. The uniform collapse hypothesis is tested by a one-sided Wilcoxon signed-rank test comparing μ_{pre} against μ_{inter} across all 54 patients, with Benjamini-Hochberg FDR correction.

Lobe enrichment. MI phenotype is cross-tabulated against the target lobe field from participants.tsv. Odds ratios (ORs) and two-sided p -values are computed via Fisher’s exact test.

3.3. Phase 2: MI-Weighted Graph Topology

Dynamic MI network. At each window w_t , construct a weighted undirected graph $G(t) = (V, E, W(t))$, where $V = \{1, \dots, 16\}$ are the channels and $W_{ij}(t) = \hat{I}(x_i^{(w_t)}; x_j^{(w_t)})$ is the MI-based edge weight.

Proportional thresholding. Retain the top 20% of edges by weight at each time step, yielding a thresholded graph $G_\tau(t)$. Proportional thresholding keeps the number of retained edges constant across time, separating topology changes from absolute MI level changes.

Network metrics. Compute at each window:

$$\rho(t) = \frac{|E_\tau(t)|}{N(N-1)/2} \quad (\text{network density}) \quad (4)$$

$$\bar{d}(t) = \frac{1}{N} \sum_i d_i(t) \quad (\text{mean degree}) \quad (5)$$

$$C(t) = \frac{1}{N} \sum_i C_i(t) \quad (\text{global clustering coefficient}) \quad (6)$$

252 Clustering is computed following Watts and Strogatz (1998) [14] via the Brain Connectivity Toolbox [15]. 297

253 Fragmentation classification. For each patient, compute the mean of each metric in the 60-second preictal window (m_{pre}) and the interictal epoch (m_{inter}). A patient is classified as fragmentation if all three metrics decrease significantly (each one-sided Wilcoxon $p < 0.05$), and as integration if all three increase significantly. 303

260 Saturated networks. Patients with interictal density $\rho_{\text{inter}} > 0.98$ are excluded from Phase 2, because the fragmentation concept does not apply to a fully connected interictal baseline. 307

264 3.4. Phase 3: MI versus Signal Variance 309

265 Global variance signal. Define the global mean signal variance at each window: 311

$$V(t) = \frac{1}{N} \sum_i \text{Var}(x_i^{(w_t)}) \quad (7) \quad 313$$

267 $V(t)$ represents the information available to a standard amplitude-based clinical alarm system. 317

269 MI-variance classification in Type A patients. For each of the 16 Type A patients ($d_\Phi > 0.5$), classify the preictal window as: (a) invisible precursor ($d_\Phi > 0.3$ and $d_V < 0$, i.e., MI collapse while variance increases); (b) co-collapse ($d_\Phi > 0.3$ and $d_V > 0.3$); or (c) other. 318

274 Independence validation. Compute the Pearson correlation $r(\Phi, V)$ separately in the interictal epoch for Type A patients, Type B patients, and the full cohort. The pre-specified threshold $|r| < 0.4$ constitutes independence of the two signals. 322

279 The original Phase 3 design specified an advance-time metric $\Delta\tau = t_{\text{collapse},V} - t_{\text{collapse},\Phi}$. This metric was not computable in the HUP dataset: the 120-second ictal clips begin within the preictal period, so the threshold-crossing collapse detector identifies only 5 MI collapses and 4 variance rises across 54 patients, with no overlap. Phase 3 findings are therefore reported as MI-variance divergence rather than MI temporal advance (see Section 5.4 for discussion). 330

288 3.5. Phase 4: Transfer Entropy SOZ Identification 331

289 TE computation. For each channel pair (i, j) , compute Transfer Entropy $\text{TE}_{i \rightarrow j}$ and $\text{TE}_{j \rightarrow i}$ during the 10-second ictal onset window $[t_s, t_s + 10]$ using the Schreiber estimator [16] with embedding dimension $k = l = 3$ and lag set to the first zero-crossing of each channel’s autocorrelation function, bounded to $[1, 50]$ ms. 336

295 TE asymmetry index. For each channel i , compute: 337

$$\text{TE}_{i \rightarrow \text{rest}} = \frac{1}{N-1} \sum_{j \neq i} \text{TE}_{i \rightarrow j} \quad (\text{mean outgoing TE}) \quad (8) \quad 339$$

$$A_i = \text{TE}_{i \rightarrow \text{rest}} - \text{TE}_{\text{rest} \rightarrow i} \quad (\text{asymmetry index}) \quad (9) \quad 341$$

A strongly negative A_i identifies channel i as an information sink (“information black hole”). Channels are ranked by A_i in ascending order, with the most sink-like channels ranked first as predicted SOZ candidates.

SOZ proxy labels. The public HUP BIDS release does not include per-electrode SOZ annotation columns in *_electrodes.tsv. Per-electrode SOZ proxy labels were derived by mapping each electrode’s MNI coordinates to a lobe via a simplified anatomical atlas and assigning SOZ membership to electrodes whose assigned lobe matched the patient’s target lobe in participants.tsv. Five patients received no labels because their target lobe (INSULAR, MFL) had no atlas match; 50 patients were labelled. Label sources: mni_TEMPORAL ($n = 23$), mni_MTL ($n = 14$), mni_FRONTAL ($n = 10$), mni_FP ($n = 1$), mni_PARIETAL ($n = 1$), mni_MFL ($n = 1$).

Performance evaluation. For each patient, we computed the area under the ROC curve (AUC) treating A_i as a continuous SOZ score, and precision at top- k for $k \in \{1, 3\}$. All AUC values are lower bounds on true performance, because the proxy labels include non-SOZ electrodes within the target lobe.

4. Results

4.1. Phase 1: Uniform Collapse Rejected; Bimodal Bifurcation Revealed

The group-level Wilcoxon signed-rank test comparing μ_{pre} against μ_{inter} across all 54 patients was not significant ($p = 0.92$, mean $\Delta = -168\%$, mean Cohen’s $d = -4.39$). The uniform MI collapse hypothesis is rejected.

The rejection is not a null result: the per-patient distribution of Cohen’s d values is strongly bimodal rather than centred near zero (Fig. ??c). Classifying patients by effect size threshold reveals:

- Type A (MI collapse): $n \approx 20$ patients (49%), mean $d = +0.99$, range $[+0.32, +2.88]$. Preictal $\Phi(t)$ is consistently lower than the interictal baseline.
- Type B (MI amplification): $n \approx 15$ patients (37%), mean $d = -2.63$, range $[-53.1, -0.32]$. Preictal $\Phi(t)$ exceeds the interictal baseline, in extreme cases by an order of magnitude.
- Indeterminate: $n \approx 6$ patients (14%), $|d| \leq 0.3$.

Table 1 reports the lobe enrichment statistics. Pooling 20 Type A patients (positive d) with 15 Type B patients (negative d) in a Wilcoxon test produces $p = 0.92$ by construction: the effects cancel. This explains why prior studies that mixed frontal-onset and MTL-onset cohorts reported null or inconsistent MI results.

Table 1: Phase 1 per-phenotype summary. Frontal-onset patients are enriched 7.5-fold in Type A (OR = 7.54, $p = 0.10$); MTL-onset patients are depleted from Type A (OR = 0.26, $p = 0.13$). Both associations are in the direction predicted by lobe-specific ictogenesis mechanisms.

Phenotype	n	Mean d	$\Phi_{\text{pre}}/\Phi_{\text{inter}}$	Lobe OR	p (Fisher)
Type A (collapse)	≈ 20	+0.99	0.73	Frontal: 7.54	0.10
Type B (amplification)	≈ 15	-2.63	1.41	MTL: 0.26	0.13
Indeterminate	≈ 6	-0.01	1.01	—	—

4.2. Phase 2: Graph Topology Independently Mirrors the MI Bifurcation

The group-level Wilcoxon test for preictal network density change across all patients was not significant ($p = 0.1$) for density, $p = 0.11$ for degree, $p = 0.20$ for clustering. Twelve of 54 patients were excluded from Phase 2 because their interictal networks were fully saturated ($\rho_{\text{inter}} > 0.98$), leaving 42 for graph analysis.

Among these 42 patients, the patient-level classification reveals a near-symmetric bimodal split (Fig. ??):

- Network fragmentation: 16 patients show simultaneous significant decreases in all three metrics (each $p < 0.05$). Mean density drop = 47%; Cohen’s d for density ranges from 1.2 to 5.4.
- Network integration: 15 patients show simultaneous significant increases in all three metrics. Mean density increase = 38%.
- Unclassified: 11 patients show mixed or non-significant metric changes.

The 16/31 fragmentation split mirrors the approximately 20/35 Type A split in Phase 1, and the median Cohen’s d for network density across all 42 patients is 0.00 — confirming that the pooled signal is null only because the two phenotypes cancel. Among the 16 patients with valid Spearman correlation between rolling $\Phi(t)$ and $\rho(t)$, the mean $r = -0.60$, confirming that MI signal level and MI-weighted graph density capture the same underlying network dynamics from complementary perspectives. Critically, a MI estimator artefact would not independently produce the same bimodal split in a graph topology measure derived by thresholding rather than averaging. The Phase 2 result constitutes independent confirmation that the Phase 1 bifurcation is a genuine network phenomenon.

4.3. Phase 3: MI and Variance Diverge in Frontal-Onset Patients

Among the 16 Type A patients ($d_{\Phi} > 0.5$), MI and variance diverge in 6 of 16 cases (38%): MI collapses while variance simultaneously increases (Table 2). For these patients, a variance-based clinical alarm would report no preictal signal — or a spurious upward crossing — while MI correctly identifies network information loss.

The interictal Pearson correlation $r(\Phi, V)$ confirms signal independence (Table 3). Type A (frontal-onset) patients show mean $r = 0.362$, below the pre-specified 0.4

independence threshold. Type B (MTL-onset) patients show $r = 0.497$, above 0.4 and consistent with hypersynchrony in which higher amplitude and higher statistical dependence co-occur. The full-cohort mean $r = 0.493$ exceeds 0.4 because Type B patients dominate the cohort mean. This confirms that MI and variance are qualitatively distinct signals for precisely the patient subgroup in whom the information bottleneck dynamic operates.

4.4. Phase 4: TE Asymmetry — Partial Signal Under Proxy Labels

Phase 4 processed 55 patients, of whom 50 received MNI proxy SOZ labels. The aggregate AUC does not confirm the H4 hypothesis (AUC > 0.75): mean AUC = 0.502 (SD = 0.232), not significantly above chance.

The lobe-specific pattern is consistent with the Phase 1–3 results (Table 4). MTL-targeted patients show mean AUC = 0.584, with 5 of 14 (36%) exceeding 0.75. The four highest-performing patients — HUP135 (AUC = 1.00), HUP190 (AUC = 1.00), HUP163 (AUC = 0.91), and HUP187 (AUC = 0.84) — are all MTL-targeted. Mean precision at top-1 was 0.28 and at top-3 was 0.41, both above the chance level of approximately 0.15–0.20 for typical SOZ fractions among 16 channels. Frontal-targeted patients show mean AUC = 0.406, 1 of 10 exceeding 0.75, consistent with the distributed anatomy of frontal SOZ that reduces the spatial specificity of the MNI proxy labels.

The mean AUC of 0.50 is a lower bound, not a performance ceiling, for three reasons: the proxy labels assign SOZ membership to all electrodes in the target lobe, inflating the SOZ set with non-SOZ channels; the coarse atlas boundaries cannot resolve fine-grained sulcal SOZ anatomy; and five patients with INSULAR or MFL targets received no labels at all. The partial signal in MTL patients (9 of 50 patients exceeding AUC 0.75; precision@1 = 0.28 above chance) constitutes preliminary evidence for the information black hole hypothesis sufficient to motivate obtaining exact per-electrode SOZ annotations from the HUP clinical team.

5. Discussion

5.1. The Bifurcation Resolves a Longstanding Literature Inconsistency

The central finding — that preictal MI direction is lobe-specific rather than universal — directly explains why

Table 2: MI-variance pattern in Type A patients ($n = 16$). 6 of 16 (38%) show an invisible precursor: MI collapses while variance increases, a preictal signature undetectable by amplitude-based monitoring.

Pattern	Description	n	%
Invisible precursor	$d_\Phi > 0.3$, $d_V < 0$: MI collapses, V increases	6	38
Co-collapse	Both $d_\Phi > 0.3$ and $d_V > 0.3$	9	56
Neither threshold	—	1	6

Table 3: Interictal Pearson $r(\Phi, V)$ by phenotype. Type A (frontal-onset) patients fall below the 0.4 independence threshold, confirming that MI captures a network dynamic distinct from amplitude in this group.

Group	n	Mean $r(\Phi, V)$
Type A (collapse)	16	0.362
Type B (amplification)	15	0.497
Full cohort	54	0.493

MI-based seizure studies have disagreed for two decades. Mormann et al. (2007) [3] noted that preictal synchrony decreases are observed predominantly in temporal lobe epilepsy cohorts while frontal and extratemporal results are inconsistent. From the present findings, this is expected: a cohort enriched for MTL patients will observe MI amplification (the Type B pattern), a cohort enriched for frontal patients will observe MI collapse (Type A), and a mixed cohort will observe a null result. The lobe enrichment odds ratios (frontal OR = 7.54 in Type A; MTL OR = 0.26 in Type A) predict exactly which cohort composition produces which directional result.

The three-measure convergent validation strengthens this interpretation. The same bimodal split appears in MI signal level, graph topology, and MI-variance decorrelation — three measures derived by distinct mathematical operations (averaging, thresholding, correlation). In contrast to Schindler et al. (2008) [4], whose linear correlation measure identified group-level changes without resolving the directionality split, the MI-based measures separate the two phenotypes cleanly. Convergence across three measures provides strong evidence that the bifurcation is a genuine property of the underlying neural networks, not a Kraskov estimator artefact.

5.2. Biological Interpretation

Type A patients (frontal onset) exhibit the information bottleneck compression dynamic [17]. Frontal networks are large, distributed, and functionally heterogeneous in the healthy interictal state, with many channel pairs maintaining low MI because they process distinct information streams. As the focal SOZ drives pathological entrainment preictally, this distributed processing collapses into synchronous co-activation, and pairwise MI falls network-wide — consistent with the preictal derecruitment of distributed frontal cortex described in the ictogenesis literature [18].

Type B patients (MTL onset) exhibit the mirror dynamic. The hippocampal-entorhinal circuit is extensively recurrently connected in the healthy baseline [19], producing above-average interictal pairwise MI between adjacent depth contacts. The preictal transition in MTL epilepsy involves progressive entrainment of CA1-CA3-entorhinal circuits that raises pairwise MI further above the interictal baseline — the hypersynchrony mechanism consistent with Mormann et al.’s (2007) [3] temporal lobe findings and with the network-level correlations reported by Schindler et al. (2008) [4] in temporal seizures.

5.3. Clinical Implications

A universal MI threshold for seizure detection will fail approximately half the population. For Type A patients, the correct detection rule uses a downward threshold: $\Phi(t) < \mu_{\text{inter}} - k\sigma_{\text{inter}}$ signals the preictal state. For Type B patients, an upward threshold: $\Phi(t) > \mu_{\text{inter}} + k\sigma_{\text{inter}}$. Both thresholds can be calibrated from the interictal baseline that every implanted BCI accumulates continuously — no seizure labels are required. A practical MI phenotype classifier would assign each patient to Type A or Type B using the sign of Cohen’s d computed on an initial interictal-to-preictal calibration window, then deploy the appropriate threshold polarity.

The invisible precursor finding carries separate clinical weight. For 6 of 16 Type A patients (38%), MI collapse occurs while variance simultaneously increases. An amplitude-based alarm — the current approach in responsive neurostimulation devices [2] — would fire in the wrong direction or not at all. The MI advantage in these patients is not quantitative (MI detects earlier than variance) but qualitative: MI detects a network-level reorganisation that is physically invisible to power monitoring.

5.4. Limitations

SOZ proxy labels. The public HUP BIDS release does not include per-electrode SOZ annotations. The MNI proxy labels used in Phase 4 assign SOZ membership to all electrodes within the target lobe, inflating the SOZ set and suppressing AUC. This means the Phase 4 AUC of 0.50 is a lower bound; its distance from the target AUC of 0.75 cannot be attributed to the TE asymmetry signal alone. True evaluation of H4 requires exact per-electrode annotations from the HUP clinical team or validation on a second dataset with published SOZ labels [20].

Recording design and Phase 3. The 120-second preictal clips prevent evaluation of the MI temporal advance over

Table 4: Phase 4 AUC by MNI proxy label source. MTL-targeted patients show the strongest TE asymmetry signal (mean AUC = 0.584, 36% above 0.75), consistent with their compact hippocampal-entorhinal seizure network. All AUC values are lower bounds against exact per-electrode SOZ annotations.

Lobe source	n	Mean AUC	AUC > 0.75	Precision@1
MTL	14	0.584	5 (36%)	—
Temporal	23	0.481	2 (9%)	—
Frontal	10	0.406	1 (10%)	—
All labelled	50	0.502	9 (18%)	0.28

variance, because the clips begin within the preictal period. Datasets with continuous long-term recordings — e.g., the MNI Open iEEG corpus — would enable the advance-time analysis as originally designed.

Lobe enrichment statistics. The frontal OR = 7.54 ($p = 0.10$) and MTL OR = 0.26 ($p = 0.13$) are directionally consistent with the predicted mechanisms but do not reach conventional significance. The classified sample ($n \approx 35$) is underpowered for Fisher’s exact test. Formal significance requires a larger cohort or reclassification using clinical SOZ lobe rather than the participants.tsv surgical target lobe, which is an approximation.

Single site, single dataset. All analyses used a single-institution cohort. External replication on an independent iEEG dataset is required before the bifurcation finding can be claimed to generalise across recording centres, electrode configurations, and patient populations.

6. Conclusion

Mutual information between iEEG channels has been proposed as a preictal seizure biomarker for over two decades, yet published studies have consistently disagreed on whether MI increases or decreases before seizure onset, and no prior work identified the inconsistency as a systematic patient-level phenomenon.

We tested the uniform MI collapse hypothesis in 54 patients from the publicly available HUP iEEG dataset using four zero-parameter information-theoretic measures. The hypothesis was rejected ($p = 0.92$), and patient-level analysis revealed that the rejection arose from two equally prevalent, biologically opposed dynamics: frontal-onset patients exhibited preictal MI collapse (Cohen’s $d \approx +0.99$), consistent with information bottleneck compression in distributed frontal networks, while MTL-onset patients exhibited preictal MI amplification ($d \approx -2.63$), consistent with hippocampal-entorhinal hypersynchrony. This bifurcation was confirmed independently by MI-weighted graph topology (16 fragmentation vs. 15 integration; $r = -0.60$ between MI and network density) and by MI-variance divergence (frontal-onset interictal $r = 0.362$ vs. MTL-onset $r = 0.497$; 6 of 16 frontal-onset patients showed an invisible preictal precursor undetectable by amplitude monitoring).

The bifurcation — not the collapse or amplification individually — is the replicable finding. Cohorts enriched for frontal-onset seizures will observe MI collapse; cohorts

enriched for MTL-onset seizures will observe MI amplification; mixed cohorts will observe a null result, explaining the prior literature’s inconsistency. The MI directionality phenotype is a parameter-free, biologically grounded biomarker of seizure network type: the correct detection threshold polarity can be assigned from an interictal calibration window alone, without any labelled seizure data, opening a direct path toward personalised, zero-parameter closed-loop seizure detection.

Data and Code Availability

The HUP iEEG dataset is publicly available via NE-MAR (OpenNeuro ID: ds004100) and Pennsieve Discover (dataset 179); no data access agreement is required. All analysis code (preprocessing, Kraskov MI estimator, rolling $\Phi(t)$ computation, graph metrics, variance comparator, TE asymmetry index) is published as Kaggle dataset mdariffaysalnayem/ieeg-ib-core and will be archived to a GitHub repository with a Zenodo DOI at time of submission. All phases run on standard CPU hardware; no GPU is required.

Declarations

Competing interests. The authors declare no competing interests.

Funding. [To be completed.]

Ethics. All data are de-identified and publicly available; no additional ethics approval was required.

Appendix A. MI Estimator Sensitivity Analysis

The Kraskov estimator has known finite-sample bias that increases as window length decreases. The 5-second window at 500 Hz provides 2,500 samples per channel pair — adequate for $k = 5$ nearest neighbours — and the bias is consistent across the recording, affecting Φ_{inter} and Φ_{pre} equally. Supplementary Figure S2 will report the per-patient Cohen’s d for window lengths from 1 to 30 seconds and $k \in \{3, 5, 7, 10\}$, demonstrating stability of the bifurcation classification across all tested parameters.

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